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MILITARY SPECIFICATION

REGULATOR, CONSTANT CURRENT, TAXIWAY LIGHTING, TYPE M-1

1. SCOPE

1.1 This specification covers one type of taxiway lighting constant current regulator, designated Type M-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Box, Wood, Nailed And Lock-Corner

Military

MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-T-27535	Transformer, Power Isolation, Series Circuit, Airport Lighting, General Specification For

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Specifications And Standards Order Of Precedence For The Selection Of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44, New York NY 10017.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard parts. With the exception in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

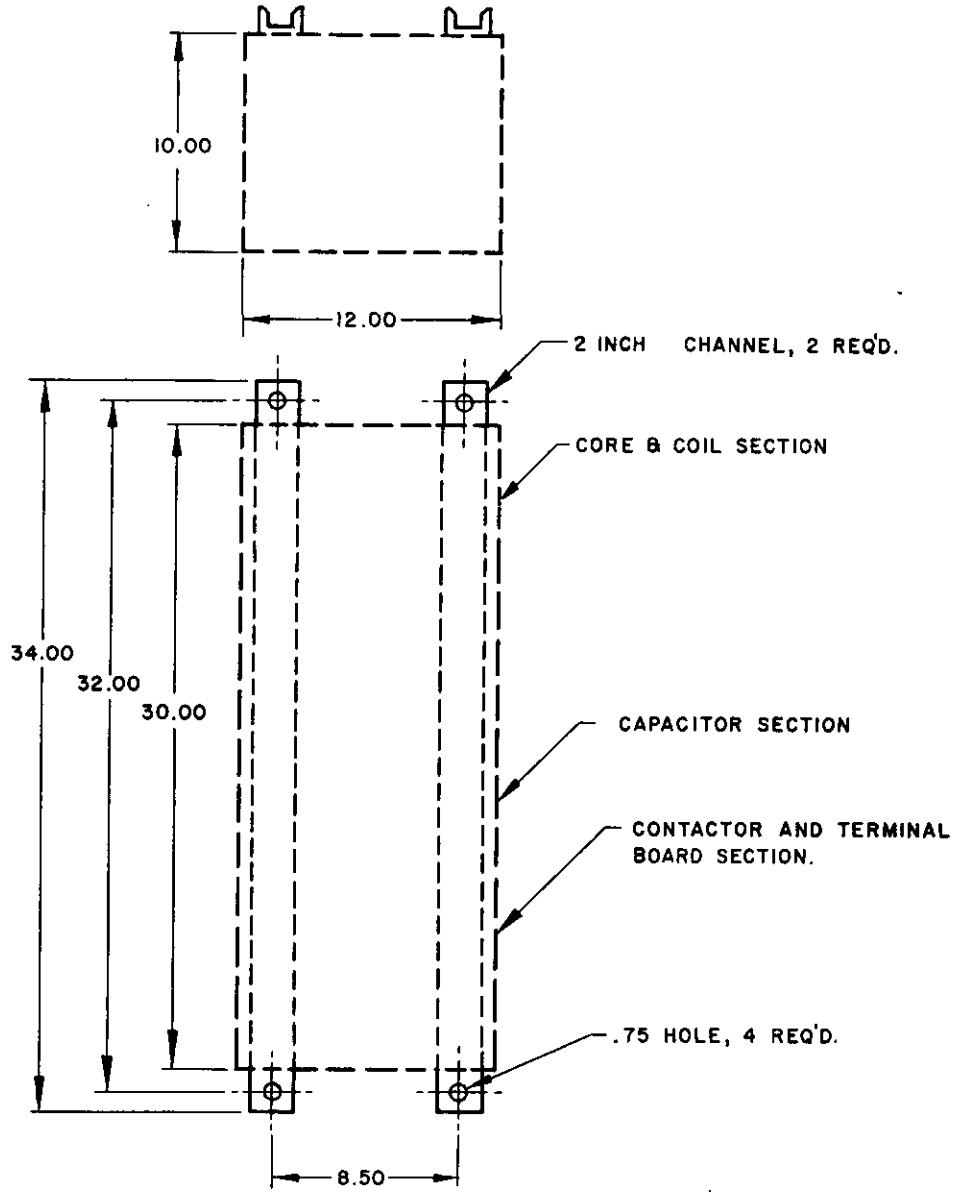
3.3 Materials.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to salt spray or atmospheric conditions likely to be met in storage or normal service.

3.4 Design and construction. The regulator shall consist essentially of a resonant circuit, constant current network, and a control circuit mounted in a steel case. The regulator shall be designed for operation on a 210, 220, 230, 240, and 250 volt, 2-wire, single-phase, 60-Hertz (Hz), constant voltage source and shall be capable of supplying a 6.6 amperage (amp) series lighting load of 1,500 volt ampere. The regulator shall be of the air-cooled, indoor type and shall be designed for wall or rack mounting as shown on Figure 1.

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED.
TOLERANCE ON:

ANGLES
DECIMALS
FRACTIONS

FIGURE I. MOUNTING

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3.4.1 Resonant circuit network. The constant current network shall have no moving parts and shall utilize a resonant circuit to produce a constant output current of the values specified herein when the proper voltages are supplied. Mineral oil capacitors, or other capacitors affected very little by extreme temperatures, shall be used to prevent output current changes with temperature. The rating shall be plainly marked on the case of each capacitor, and capacitors shall be readily accessible for test and replacement.

3.4.1.1 Any suitable control and constant current circuits may be utilized provided that all performance requirements specified herein are met and the circuits are approved by the procuring activity prior to the fabrication of any regulator.

3.4.2 Input voltage taps. Taps shall be provided so that the regulator will supply the specified output current at input voltages of 210, 220, 230, 240, and 250 volts. The taps may be connected either to terminals or to a tap-changing switch.

3.4.3 Contactor. The regulator shall be provided with a 2-pole, size 0, non-inclosed magnetic contactor, without overload protection, manufactured in accordance with National Electrical Manufacturer's Association requirements. The contractor shall be mounted inside the regulator case. The operating coil of the contactor shall be rated at 110 volts, 60 Hz.

3.4.4 Terminal boards. Pressure type terminals rated at 30 amps, 600 volts shall be provided within the regulator case for the input leads, output leads, and 110 volts control circuit, and shall be labeled "input," "output," and "control," respectively.

3.4.5 Case. All parts of the regulator shall be inclosed in a sheet steel case, except that the case may be divided into 2 or more sections, with the sides of the core exposed between sections. Ventilating louvers may be used. The shape of the case is optional, provided the mounting dimensions are in accordance with Figure 1 and no part of the case extends outside the rectangular envelope inclosed by the dotted lines shown on Figure 1. The case shall have a removable cover plate to provide access to terminals, capacitors, and control devices. The control and terminal section shall be provided with 2 knockouts for 1-inch conduits on the bottom, back, and both sides.

3.4.6 Open circuit. The open-circuit secondary voltage of the regulator shall not exceed 200 percent of the rated load voltage. The regulator shall be designed to withstand, without damage, continuous operation with secondary voltage open circuited. An open circuit protective device need not be provided.

3.4.6.1 Optional design. The regulator need not be designed to operate indefinitely with the secondary voltage open circuited if an automatic open circuit protective device is provided. The device shall open the main contactor within 1 second after failure occurs; the contactor shall remain open until resetting is accomplished by de-energizing the control circuit.

3.5 Performance. The regulator shall be capable of operating satisfactorily at any ambient temperature from -55° to +55° Centigrade (C).

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3.5.1 Output rating. The regulator shall be capable of delivering its specified output at 6.6 amp to a unit power factor load when the supply voltage is 230 volts. The output current at room temperature, approximately 20°C, with any unity power factor load from no load (short circuit) to full load shall be 6.6 amp when 60-Hz power of 210, 220, 230, 240, or 250 volts is applied and the corresponding tap utilized. The permissible output current tolerance shall be ±0.10 amp at full load and shall gradually increase to ±0.20 amp at short circuit. Full load of this regulator is 34.4 ohms.

3.5.1.1 When regulator is connected to a 230 volt, 60-Hz source and operated at room temperature, approximately 20°C, at full load consisting of lamps connected to the output circuit through isolation transformers which meet the electrical requirements of MIL-T-27535, the output current shall be 6.6 ±0.4 amp when the secondaries of the isolating transformers supplying 30 percent of the load are opened.

3.5.2 Temperature rise. The temperature rise in continuous operation for any load from short circuit to full load shall not exceed 55°C for short circuit to full load shall not exceed 55°C for class A insulation, 80°C for class B insulation, or 140°C for class H insulation.

3.5.3 Efficiency and power factor. The overall efficiency, neglecting the contactor coil, and primary power factor shall be not less than the values specified in Table I when the regulator is operated on 60-Hz, 230 volt power (utilizing the proper tap), with a unity power factor load.

TABLE I

OVERALL EFFICIENCIES AND PRIMARY POWER FACTORS

Percent of Rated KW Output	Overall Efficiency in Percent (Minimum)	Primary Power Factor in Percent (Minimum)
25	70	77
50	80	85
100	85	90

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

3.7 Wiring. The regulator assembly shall be completely wired so that only attachment of wires to terminals will be required at installation.

3.8 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the regulator, they shall be held or limited accordingly.

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3.9 Finishes. The interior and exterior of the case, and other exposed parts, shall be finished with three coats of weatherproof gray paint. Any standard, commercial gray paint may be used.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.11 Workmanship.

3.11.1 General. The regulator, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws, and bolts; et cetera.

3.11.2 Screw assemblies . Assembly screws and bolts shall be tight. The word "tight" shall mean that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.11.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.11.4 Cleaning. The regulator shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of the regulator shall be classified as follows:

a. Preproduction tests (see 4.4)

b. Quality conformance tests (see 4.5)

4.3 Test conditions. Unless otherwise specified, the tests specified herein shall be conducted at 60 Hz and at room temperature.

4.4 Preproduction testing.

* 4.4.1 Preproduction test samples. The preproduction test samples shall consist of models representative of the production equipment.

* 4.4.2 Preproduction tests. Preproduction tests shall consist of all tests described under 4.6.

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* 4.5 Quality conformance tests. Quality conformance tests shall consist of individual tests.

* 4.5.1 Individual tests. Each regulator shall be subjected to the following tests as described under 4.6.

a. Examination of product

b. Dielectric

c. Output rating

d. Contactor

4.6 Test methods.

4.6.1 Examination of product. The regulator shall be inspected to determine compliance with this specification with respect to material, workmanship, and marking.

4.6.2 Dielectric. The circuits shall be subjected to the following root mean square (rms) voltages at commercial frequencies for 1 minute. The circuits shall withstand this test without failure.

a. Control circuit to ground, 1,000 volts.

b. Input circuit to ground, 1,500 volts. (Control circuit shall be grounded, output circuit shall be connected to input circuit.)

4.6.3 Performance. The regulator, or any part thereof, shall be subjected to the following tests to determine proper operation and compliance with the requirements specified herein.

4.6.3.1 Overall efficiency and primary power factor. With 230 volts connected to the 230 volt tap and a lamp or resistance load connected to the secondary, the overall efficiency and primary power factor tested at the loads specified in Table I shall be as specified herein.

4.6.3.2 Temperature rise. The temperature rise shall be measured at the maximum temperature rise conditions by any method satisfactory to the procuring activity to determine if the temperature rise is within the limits specified in 3.5.2.

4.6.3.3 Output rating. The output current shall, when tested at all five input voltage taps with the output short circuited through an ammeter and with rated unity power factor load, be in accordance with the value specified in 3.5.1.

4.6.3.4 Contactor. The contactor shall be tested for proper operation in accordance with 3.4.3.

4.6.4 Environmental tests. The regulator shall be subjected to the following tests conducted in accordance with MIL-E-5272.

a. High temperature, except that the temperature shall be +55° ±2°C.

b. Low temperature, procedure I, except that the temperature shall be -55° ±2°C.

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4.6.5 Open circuit voltage. The open circuit voltage shall be checked to determine that it does not exceed 200 percent of rated load.

* 4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and sotrage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

* 5.1.1 Level A. Regulators shall be preserved and packaged one each, Method IIA of MIL-P-116. The unit container shall be the shipping container and shall conform to the requirements of 5.2.

* 5.1.2 Level C. Regulators shall be preserved and packaged in a manner which affords adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

* 5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. The regulator preserved and packaged as specified in 5.1 shall be packed in a nailed wood shipping container conforming to PPP-B-621 class 2. Closure and strapping shall be in accordance with the appendix to PPP-B-621.

* 5.2.2 Level B. Level B shall be the same as Level A except the shipping container shall conform to class 1.

* 5.2.3 Level C. Regulators preserved and packaged as specified in 5.1 shall be packed in a manner which affords adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

* 5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The Type M-1 regulator covered by this specification is intended to supply constant current to a series taxiway lighting circuit.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable level of preservation, packaging and packing (see 5.1 and 5.2).

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* 6.2.1 Preproduction tests. One regulator will be required as a preproduction sample to be subjected to the preproduction tests to determine compliance with the requirements of this specification.

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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